

RETINAL EDGE TRIAGE

Explainable AI-Assisted Diabetic Retinopathy Screening for Rural Healthcare
Official Clinical Triage Report | Generated: 04-Sep-2026 14:49 UTC

Patient ID: RET-2026-000027	Full Name: Test Screening Patient	Age / Gender: 52 / Female
Screening ID: SCR-2026-000026	Location: Test Rural PHC	Diabetes Type / Duration: Type 2 (6 years)

1. Retinal Image Acquisition & Quality Gate

Eye Evaluated: Right Eye	Image Source: Portable Fundus Camera	Quality Score: 67.3%	Quality Status: BORDERLINE
Focus / Blur: 31.2%	Illumination: 52.2%	Vessel Visibility: 79.7%	Field of View: 91.5%

2. AI Screening Assessment (APTOS 2019 & IDRiD Models)

DR Severity Grade:	Grade 2 — Moderate DR	Referable DR:	YES (Action Required)
AI Confidence:	96.6%	DME Risk Status:	Possible DME detected (Multiple hard exudates identified within parafoveal region, indicating potential macular edema risk.)
Detected Lesions (IDRiD):	Microaneurysms: 8 Hemorrhages: 6 Hard Exudates: 11 Soft Exudates: 1	Model Version:	Enhanced Clinical Ensemble v2.0 (Lesion-Grounded Fusion + TTA)

3. Human-in-the-Loop Ophthalmologist Adjudication

Doctor Decision:	ACCEPTED	Reviewing Doctor:	Dr. Ophthalmologist
Final Grade Assigned:	Grade 2	Override Justification:	None
Clinical Notes:	AI analysis matches clinical presentation.	HCW Instructions:	None

4. Risk Triage & Clinical Referral Plan

Risk Category:	HIGH RISK	Referral Priority:	URGENT
Referral ID:	REF-2026-00014	Recommended Facility:	District Eye Hospital
Triage Reason:	High risk DR (Grade 2) with concomitant DME suspicion.	Follow-up Schedule:	1 Months

STATUTORY CLINICAL NOTICE: This report is generated by an artificial intelligence screening triage prototype (Retinal Edge Triage). AI predictions DO NOT constitute a definitive medical diagnosis. Final clinical interpretation and management must always be performed by a qualified

ophthalmologist. Compliant with rural screening clinical governance principles.